

QDay Prize Submission Brief

End-to-End Compilable Quantum Elliptic Curve Discrete Logarithm

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Approach

We implement **Shor’s algorithm for the Elliptic Curve Discrete Logarithm Problem (ECDLP)** using **Qrisp**, a high-level quantum programming language built on top of Qiskit. Given a generator point G and public key $P = kG$ on an elliptic curve $y^2 = x^3 + 7 \pmod{p}$, the algorithm recovers the private key k via quantum phase estimation on the group operation.

The core algorithm prepares two n -qubit QPE registers in superposition, performs $2n$ controlled EC point additions (multiplying by both G and P), applies inverse QFT, and extracts k from the measurement outcomes via continued-fraction expansion. This follows the standard Shor ECDLP construction described in our paper [1].

Techniques and Design Choices

Static circuit compilation. As detailed in [1], we realize all EC arithmetic—point addition, point doubling, Kaliski modular inversion, Montgomery modular multiplication—as compilable quantum circuits using Qrisp’s `QuantumModulus` type and in-place modular arithmetic primitives. Every operation is decomposed into elementary gates (H, CNOT, T, $T^\dagger$) at compile time.

Dynamic compilation via JAX (jasp). Beyond the static approach, we introduce **JAX-traceable dynamic compilation** through Qrisp’s `jasp` subsystem. The quantum algorithm is traced through `jax.jit`, enabling dynamic circuit generation, loop unrolling, and optimization. This is critical for scaling to large bit sizes where static unrolling would be intractable.

BigInteger support. To handle prime fields exceeding the native 64-bit JAX integer limit, we developed a `BigInteger` class—a fixed-width array of 32-bit limbs registered as a JAX pytree node. This enables compilation for arbitrarily large primes (up to 256-bit and beyond) while remaining fully JAX-traceable. All classical computation uses Python `int` values; `BigInteger` values appear only in `QuantumModulus` register construction and internal operator dispatch.

Scalable resource estimation. Every controlled EC addition in the Shor loop has *identical gate structure* regardless of the classical point—only the modulus bit-width matters. We exploit this by counting gates for a *single* controlled addition via Qrisp’s `@count_ops` decorator and computing:

$$\text{Total gates} = 2n \times (\text{single EC addition}) + \text{overhead}$$

where overhead covers Hadamard preparation, initial encoding, inverse QFT, and measurement. This makes 256-bit resource estimation feasible on standard hardware without tracing 512 full EC additions.

Requirements and Execution

- **Software:** Python 3.10+, Qrisp (development branch with PRs #495 and #505, both pending merge), JAX, NumPy, SymPy, Matplotlib.
- **Hardware:** Standard laptop/desktop. Resource estimation for 256-bit takes minutes; full simulation is limited to small curves (≤ 5 -bit) due to exponential state-space scaling.

- **Execution:** Install via `pip install -r requirements.txt`. Run `resource_estimation_all_curves.ipynb` to compile and count gates for all QDay standard curves.
- **Validation:** Unit tests (`pytest tests/ -v`) verify correctness of all quantum primitives against classical reference implementations using Qrisp’s boolean simulator.

Potential Drawbacks

- **No fault-tolerant hardware execution:** Current quantum hardware cannot execute circuits of this depth. The submission demonstrates correctness and compilation feasibility, not hardware execution.
- **No T-gate optimization:** The current compilation does not apply T-gate reduction techniques (e.g., T-factory scheduling, magic state distillation budgeting). Gate counts represent unoptimized decompositions.
- **Qrisp dependency on development branches:** The implementation requires two PRs not yet merged into Qrisp’s main branch.
- **Static point encoding:** The current approach encodes classical EC points statically. Windowed or semi-classical QFT variants could reduce qubit count but are not yet implemented.

References

- [1] D. Polimeni, R. Seidel. *End-to-end compilable implementation of quantum elliptic curve logarithm in Qrisp*. arXiv:2501.10228 (2025).